02/08/2026

Code-E

Aakash
Medical | IIT-JEE | Foundations

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MM : 720 **AIATS Prime for NEET-2027(XII Passed)_Unit Test-01** **Time : 180 Min.**

Topics Covered:
Physics : Units & Measurements, Motion in a Straight Line
Chemistry : Structure of Atom, Some Basic Concepts of Chemistry
Botany : Cell: The Unit of Life, Cell Cycle and Cell Division
Zoology : Structural Organisation in Animals - Animal Tissues, Biomolecules

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

1. The average of time periods of a simple pendulum is 10.00 s. Which of the following measurements is the most accurate?

(1) 9.95 s

(2) 10.10 s

(3) 9.50 s

(4) 10.25 s

2. The number of significant figures for the numbers 0.03708 m, 902.60 m and 5.00×10^{-2} kg is

(1) 4, 5, 2

(2) 4, 5, 3

(3) 4, 4, 3

(4) 3, 5, 1

3. A compressive force $F = (5 \pm 0.005)$ dyne is applied on a rod of cross-sectional radius $r = (2 \pm 0.01)$ cm. The maximum percentage error in stress is

(1) 1.2%

(2) 1.1%

(3) 1.5%

(4) 0.5%

4. If velocity (V), mass (M) and time (T) are taken as fundamental quantities, then the dimensional formula of gravitational potential energy would be

(1) $[M^{-2}VT^{0}]$

(2) $[M^{2}VT^{0}]$

(3) $[MV^{-2}T^{1}]$

(4) $[MV^{2}T^{0}]$

5. Consider the following statements about significant figures:
(A) The measurement 4.700 has two significant figures.
(B) In the measurement 0.00560, only three significant figures are present.
(C) Changing the unit of measurement can alter the number of significant figures.
(D) Scientific notation removes ambiguity in determining significant figures.
The correct statement(s) is/are

(1) Both (A) and (B)

(2) Both (B) and (D)

(3) Both (C) and (D)

(4) All statements are correct

6. The SI unit of Stefan's constant is

(1) $W\,m^2\,K^4$

(2) $W\,m^{-1}\,K^{-4}$

(3) $W\,m^{-2}\,K^{-4}$

(4) $W\,m\,K^4$

7. Given below are two statements : one is labelled as **Assertion** and the other is labelled as **Reason**
Assertion (A): Measured value 1.56 is more precise than 1.5.
Reason (R): More accurate measurement will always be more precise.
In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of the (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

8. Which of the following pairs have same dimensions?

(1) Force and strain

(2) Force and stress

(3) Angular velocity and frequency

(4) Energy and strain

9. Consider following two statements.
a : Unitless quantities are dimensionless.
b : Dimensionless quantities are always unitless.
The **correct** statement(s) is/are

(1) Only (a)

(2) Only (b)

(3) Both (a) and (b)

(4) Neither (a) nor (b)

10. The velocity v (in cm/s) of a particle is given in terms of time t (in second) by the equation:
$v = at + \frac{b}{t+c}$ The dimensions of a/c are

(1) $[LT^{-2}]$

(2) $[LT^{-3}]$

(3) $[LT^2]$

(4) $[L^2T^2]$

11. The side of a cube is 2.5 meter. The volume of cube according to the appropriate significant figures is (in m^3)

(1) 15.625

(2) 16

(3) 1.5

(4) 1.6

12. If 86.45 is rounded off upto three significant figures then it should be written as

(1) 86.5

(2) 86.4

(3) 86.0

(4) 86.3

13. The least count of a stopwatch is 0.2 s, the time of 100 oscillations is found to be 200 s. The percentage error in the measurement of time is

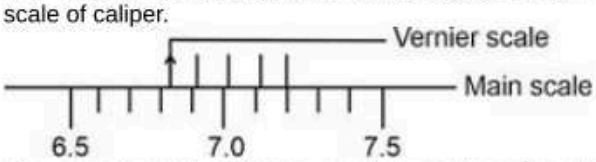
(1) 10%

(2) 0.1%

(3) 0.2%

(4) 5%

14. The diagram shows a part of vernier scale and a main scale of caliper.



The correct reading is (given least count of vernier caliper = 0.01 cm)

(1) 6.80 cm

(2) 6.84 cm

(3) 6.76 cm

(4) 7.20 cm

15. In column-I few measurements are given and in column-II, the number of significant figures are given. Match the column-I and column-II.

	Column-I		**Column-II**
a.	$0.0007\ m^2$	(i)	3
b.	2.64×10^{24} kg	(ii)	5
c.	6.320 J	(iii)	1
d.	$0.63200\ g\ cm^{-3}$	(iv)	4

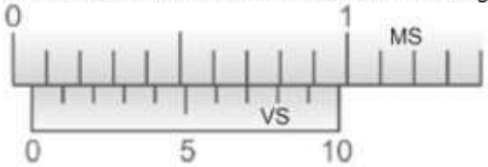
(1) a(i), b(ii), c(iv), d(iii)

(2) a(iii), b(i), c(iv), d(ii)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iv), b(ii), c(iii), d(i)

16. The figure shows a situation when jaws of vernier are touching each other and one main scale division is of 1 mm. The zero correction in measured reading will be



(1) 0.5 mm

(2) 0.7 mm

(3) –0.5 mm

(4) –0.7 mm

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17. If unit of mass, length and time each is tripled, then the unit of force will

(1) Become 2 times

(2) Remain same

(3) Become 8 times

(4) Become 9 times

18. *n*-divisions on the main scale of a vernier callipers coincide with $n + 1$ divisions on the vernier scale. If each division on the main scale is of x units, determine the least count of the instrument.

(1) $\frac{x}{n}$

(2) $\frac{x}{n+1}$

(3) $\frac{x}{n-1}$

(4) $\frac{2x}{n+1}$

19. Screw gauge has one main scale division of 1 mm and 100 circular scale divisions then the least count of screw gauge is.

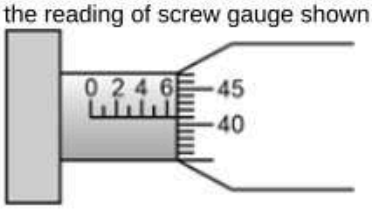
(1) 0.1 mm

(2) 0.01 mm

(3) 0.001 mm

(4) 1 mm

20. If the least count of the given micrometer is 0.01 mm, then the reading of screw gauge shown in figure is



(1) 6.41 mm

(2) 6.38 mm

(3) 6.26 mm

(4) 6.48 mm

21. The pitch of screw gauge is 1 mm and there are 100 divisions on circular scale. While measuring diameter of a wire, linear scale reads 1 mm and 39th division of circular scale coincides with reference line. The zero error is –0.05 mm. The correct diameter of wire is

(1) 1.39 mm

(2) 1.44 mm

(3) 1.35 mm

(4) 1.034 mm

22. A man walks on a train moving at 10 m/s. He walks towards the front of the train at 2 m/s w.r.t. the train. The relative speed of the man w.r.t. ground is

(1) 12 m/s

(2) 8 m/s

(3) 10 m/s

(4) 2 m/s

23. A particle initially at rest at origin starts moving along x-axis. Its acceleration-time equation is given as $a = (6t + 5)$ m/s^2. If it starts from origin, the distance covered in 4 s is

(1) 64 m

(2) 104 m

(3) 24 m

(4) 114 m

24. A particle moves along x-axis such that its velocity (v) varies with time (t) as $v = (4t - t^2)$ m/s. The distance travelled by the particle from $t = 0$ to $t = 6$ s is

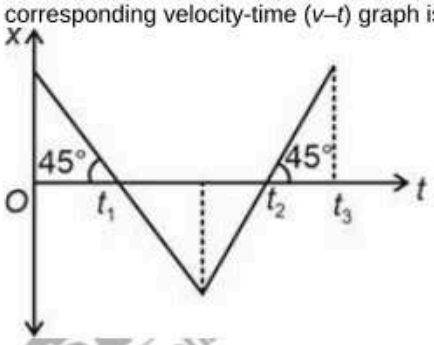
(1) 10.66 m

(2) 21.33 m

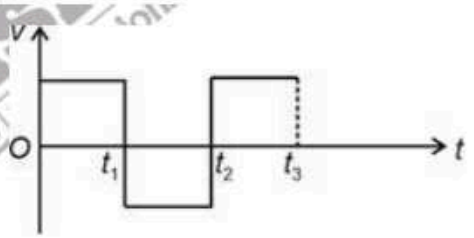
(3) 4.31 m

(4) Zero

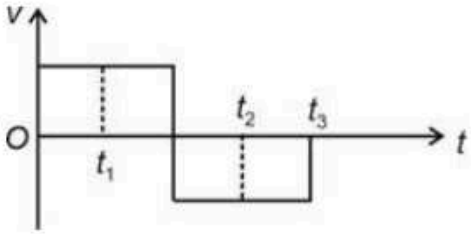
25. For the position-time (x – t) graph, the shape of corresponding velocity-time (v–t) graph is



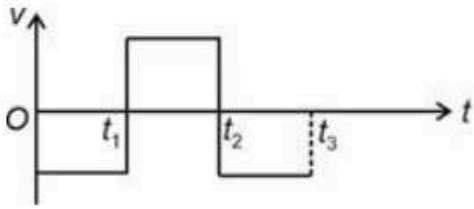
(1)



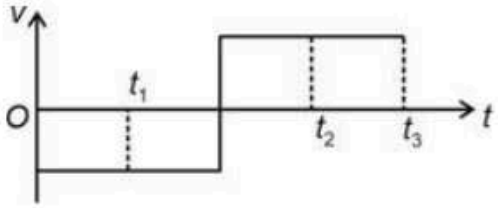
(2)



(3)



(4)



26. A car moving with a velocity of 36 kmph along a straight line is slowed down to 18 kmph in 5 s. The retardation of the car is

(1) 1 m/s^2

(2) 2 m/s^2

(3) 4 m/s^2

(4) 3 m/s^2

27. Consider the following statements:
Statement (A): If displacement of a particle is zero, the distance covered must be zero.
Statement (B): The numerical value of the ratio of magnitude of average velocity to average speed is always equal to one.
The correct statement(s) is/are

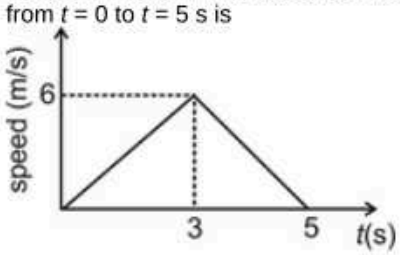
(1) Only statement (A)

(2) Only statement (B)

(3) Both (A) and (B)

(4) Neither (A) nor (B)

28. The speed-time graph of a particle moving along a curve is as shown below. The distance travelled by the particle from $t = 0$ to $t = 5$ s is



(1) 15 m

(2) 9 m

(3) 6 m

(4) 30 m

29. A stone starts falling freely under gravity. The time taken by the stone to fall through the initial height h is 4 s. The further time taken to fall through the next $2h$ height will be closest to

(1) 3 s

(2) 5 s

(3) 1 s

(4) 6 s

30. The initial velocity of a particle moving along positive x-axis is 10 ms^{-1}. If the constant acceleration $a = -2$ m s^{-2} acts on the particle, then distance travelled by the particle when it reaches to its initial position again is

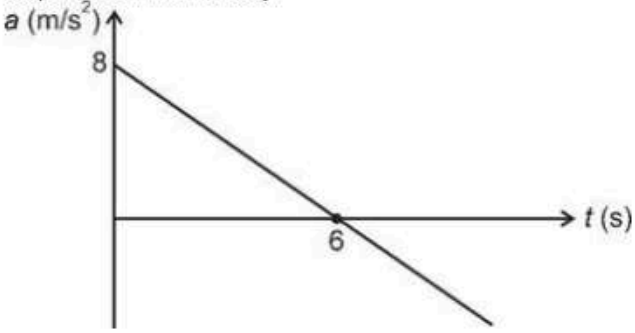
(1) 20 m

(2) 25 m

(3) 40 m

(4) 50 m

31. A particle is moving on a straight line according to given acceleration (a)-time (t) graph. At what time the particle acquires its initial velocity?



(1) $t = 9$ s

(2) $t = 16$ s

(3) $t = 12$ s

(4) $t = 15$ s

32. A particle covers half of its total distance with speed u and the rest half distance with speed v. Its average speed during the complete journey is

(1) $\frac{u+v}{2}$

(2) $\frac{2uv}{u+v}$

(3) $\frac{uv}{u+v}$

(4) $\frac{uv}{v-u}$

33. A particle is dropped from the top of a tower of height 4.9 m. The velocity of particle with which it strikes the ground is ($g = 9.8$ ms^{-2})

(1) 5 m/s

(2) 4.9 m/s

(3) 9.8 m/s

(4) 7.3 m/s

34. The area under velocity-time graph for a particle moving along x-axis, in a given time interval represents.

(1) Change in velocity

(2) Acceleration

(3) Displacement

(4) Time

35. A body starts from rest and moving with uniform acceleration a, for time t. The average velocity over time t is

(1) at

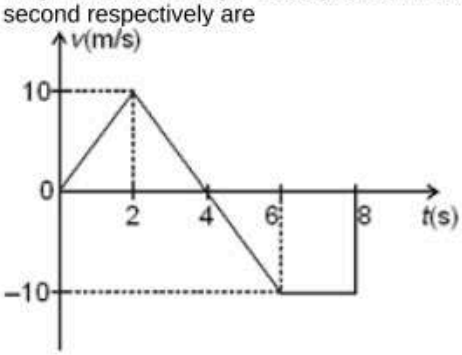
(2) $\frac{at}{2}$

(3) $\frac{at}{4}$

(4) $\frac{at^2}{2}$

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36. The given velocity-time graph shows the motion of a car along straight line path. The average speed and magnitude of average velocity of car from $t = 0$ to $t = 8$ second respectively are



(1) $\frac{25}{4}$ m/s, $\frac{-5}{4}$ m/s

(2) $\frac{25}{4}$ m/s, $\frac{5}{4}$ m/s

(3) $-\frac{25}{4}$ m/s, $\frac{-5}{4}$ m/s

(4) $\frac{-5}{4}$ m/s, $\frac{25}{4}$ m/s

37. A person moves from town *A* to town *B* with speed *v* and returns back to town *A* from town *B* along the same path with speed 3*v*. The average speed of person is

(1) $\frac{3v}{2}$

(2) $\frac{4v}{3}$

(3) Zero

(4) $\frac{8v}{3}$

38. A body falls freely from a height of 80 m. After 2 s of fall gravity disappears. Total time taken by it to reach the ground will be ($g = 10$ m/s^2)

(1) 5 s

(2) 4 s

(3) 7 s

(4) 6 s

39. A ball is thrown vertically upward with initial speed 40 m/s. The distance travelled by the ball in last second of upward journey is ($g = 10$ m/s^2)

(1) 5 m

(2) 10 m

(3) 80 m

(4) 20 m

40. A body moves along a straight line with an acceleration 3 m/s^2 for 2 s and then with an acceleration 4 m/s^2 for 3 s. What is average acceleration?

(1) 3.4 m s^{-2}

(2) 3.5 m s^{-2}

(3) 3.6 m s^{-2}

(4) 3.7 m s^{-2}

41. The position of a particle moving along x-axis, varies as $x = 4t^3 - 3t$. The acceleration of the particle, when the velocity of the particle becomes zero, is

(1) 12 m/s^2

(2) 10 m/s^2

(3) 5 m/s^2

(4) 20 m/s^2

42. A coin is released inside a lift at a height of 2 m from the floor of the lift. The height of lift compartment is 4 m. The lift is moving downward with an acceleration 5 m/s^2. The time after which the coin will strike with the lift floor is

(1) 2 s

(2) $2\sqrt{2}$ s

(3) $\frac{4}{\sqrt{3}}$ s

(4) $\frac{2}{\sqrt{5}}$ s

43. Two bodies start moving from same point along a straight line in same direction with velocities $v_1 = 16$ m/s and $v_2 = 10$ m/s simultaneously. After what time their separation becomes 60 m?

(1) 6 s

(2) 8 s

(3) 12 s

(4) 10 s

44. A train of 360 m length is going along east with 50 m/s. A bird flies at a speed 10 m/s towards west parallel to the track. The time taken by the bird to cross the train is

(1) 10 s

(2) 8 s

(3) 6 s

(4) 4 s

45. A person walked up a stalled escalator in 60 s. When standing on the same escalator, now moving, he is carried up in 30 s. How much time would it take him to walk up the moving escalator?

(1) 10 s

(2) 90 s

(3) 30 s

(4) 20 s

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CHEMISTRY

46. According to the Rutherford's nuclear model of atom
(a) Electrons and the nucleus are held together by electrostatic forces of attraction
(b) The positive charge and most of the mass of the atom was densely concentrated in extremely small region called nucleus
(c) The nucleus is surrounded by electrons that move around the nucleus with a very high speed in circular paths called orbits
The correct statement(s) is/are

(1) (b) and (c) only

(2) (a), (b) and (c)

(3) (a) and (c) only

(4) (a) only

47. Incorrect statement regarding cathode rays is

(1) The cathode rays start from cathode and move towards the anode

(2) In the absence of electric and magnetic field these rays travel in straight line

(3) The charge to mass ratio of the cathode rays depends upon the gas from which they originate

(4) Cathode rays consist of negatively charged particles

48. The longest wavelength in the Lyman series of the atomic hydrogen spectrum is [R_H = Rydberg constant]

(1) $\frac{1}{3R_H}$

(2) $\frac{4}{3R_H}$

(3) $\frac{3R_H}{4}$

(4) $\frac{R_H}{2}$

49. Isobar of $^{14}_{6}C$ is

(1) $^{12}_{6}C$

(2) $^{13}_{6}C$

(3) $^{16}_{8}O$

(4) $^{14}_{7}N$

50. Given below are two statements.
Statement I: Electromagnetic waves do not require medium and can move in vacuum.
Statement II: The wavelength of radio waves is less than that of microwave.
In the light of above statements, choose the **correct** option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

51. Neutrons were discovered by bombarding α-particles on a thin sheet of

(1) Gold

(2) Silver

(3) Beryllium

(4) Copper

52. If radius of second orbit of Li^{2+} is x Å then radius of the first orbit of He^{+} will be

(1) $3x$ Å

(2) $\frac{2}{3}x$ Å

(3) $\frac{5}{8}x$ Å

(4) $\frac{3}{8}x$ Å

53. Number of neutrons, protons and electrons present in $^{65}_{30}Zn^{2+}$ respectively are

(1) 35, 30, 30

(2) 30, 35, 30

(3) 35, 30, 28

(4) 35, 28, 30

54. An electron jumps from 4th excited state to ground state in hydrogen atom, then the number of lines obtained in UV region will be

(1) 10

(2) 6

(3) 4

(4) 3

55. A 10 W bulb emits 400 nm radiation, the rate of emission of photons per second is
(h = 6.626 ×10^{-34} Js)

(1) 2×10^{12}

(2) 5×10^{15}

(3) 5×10^{17}

(4) 2×10^{19}

56. If the wavelength of a photon is 6.02×10^{-5} cm then the energy of one mole of photons will be (h = 6.626×10^{-34} Js)

(1) 225.25 kJ

(2) 198.78 kJ

(3) 95.25 kJ

(4) 175.31 kJ

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57. When a metal is exposed to light of wavelength (λ), photoelectrons with maximum kinetic energy 2 eV are ejected from metal surface. On exposing the same metal with light of wavelength $\left(\frac{\lambda}{4}\right)$, the ejected photoelectrons have maximum kinetic energy 10 eV. The work function of metal is

(1) $\frac{2}{3}$ eV

(2) $\frac{1}{3}$ eV

(3) 2 eV

(4) 3 eV

58. The energy of 4th orbit of hydrogen atom is –82 kJ mol^{-1}, hence the energy of second orbit of Li^{2+} would be

(1) –1312 kJ mol^{-1}

(2) –328 kJ mol^{-1}

(3) –2624 kJ mol^{-1}

(4) –2952 kJ mol^{-1}

59. Given below are the two statements.
Statement I: If the de Broglie wavelength of an electron in the Bohr orbit for a hydrogen like species is equal to $3\pi a_o$($a_o$is Bohr's radius), then the value of $\frac{n}{Z}$is 1.5.
Statement II: Total number of electrons in the p subshell of sulphur atom is 10.
In light of above statements, choose the **correct** answer.

(1) Statement I is incorrect but statement II is correct

(2) Statement I is correct but statement II is incorrect

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

60. Match the following items in column I with column II and choose the correct option.

	Column I (quantum nos.)		**Column II (orbital)**
a.	n = 2, l = 1	(i)	3*p*
b.	n = 3, l = 2	(ii)	3*d*
c.	n = 1, l = 0	(iii)	2*p*
d.	n = 3, l = 1	(iv)	1*s*

(1) a(iii), b(i), c(iv), d(ii)

(2) a(iii), b(ii), c(iv), d(i)

(3) a(ii), b(i), c(iii), d(iv)

(4) a(i), b(ii), c(iii), d(iv)

61. The correct sequence of energy of orbitals of H-like species is

(1) 3*d* < 4*s* < 4*p*

(2) 4*s* < 4*p* < 3*d*

(3) 3*d* < 4*s* = 4*p*

(4) 4*s* < 3*d* < 4*p*

62. The shortest wavelength of H-atom in the Balmer series is λ_1. The longest wavelength in the Paschen series of Li^{2+} ion is

(1) $\frac{4\lambda_1}{7}$

(2) $\frac{7\lambda_1}{4}$

(3) $\frac{7}{4\lambda_1}$

(4) $\frac{4}{7\lambda_1}$

63. The orbital angular momentum of electron in 4*s* orbital of Cu atom is

(1) $\sqrt{2}\,\hbar$

(2) $2\sqrt{3}\,\hbar$

(3) Zero

(4) $\sqrt{6}\,\hbar$

64. The wavelength associated with an electron moving with a velocity of 10^5 m/s is (take: h = 6 × 10^{-34} J.s and m_e = 9 × 10^{-31} kg)

(1) 3.33 nm

(2) 6.66 nm

(3) 1.25 nm

(4) 2.5 nm

65. Two electrons occupying the same orbital are distinguished by

(1) Principal quantum number

(2) Azimuthal quantum number

(3) Magnetic quantum number

(4) Spin quantum number

66. Which of the following pairs of compounds does not follow law of multiple proportions?

(1) CO_2, CO

(2) NO_2, N_2O_5

(3) H_2O, H_2O_2

(4) SO_2, N_2O

67. Given below are two statements.
Statement I: There are $(2l+1)$ orbitals of each type in a subshell.
Statement II: Magnetic orbital quantum number, gives information about the spatial orientation of the orbital with respect to standard set of co-ordinate axis.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct.

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68. Total number of electrons in 1 mol P^{3-} ions is

(1) 18

(2) 18 N_A

(3) 15 N_A

(4) 10 N_A

69. The number of unpaired electrons in Co^{2+} ion is

(1) 4

(2) 5

(3) 6

(4) 3

70. Consider the following statements
Statement-I: Gay Lussac's Law of Gaseous Volumes : When gases combine or are produced in a chemical reaction they do so in a simple ratio by volume provided that all gases are at same temperature and pressure.
Statement-II: The volumes of $H_2(g)$ and $O_2(g)$ which combine together to form water bear a simple ratio of 2 : 1
In the light of above statements, choose the correct answer from the options given below.

(1) Statement I is false but statement II is true

(2) Both statements I and II are true

(3) Both statements I and II are false

(4) Statement I is true but statement II is false

71. Match the column I with column II.

	Column I (Prefix)		Column II (multiple)
a.	mega	(i)	10^{15}
b.	giga	(ii)	10^{12}
c.	tera	(iii)	10^{9}
d.	peta	(iv)	10^{6}

choose the most appropriate match from the option given below:

(1) a(iv), b(iii), c(ii), d(i)

(2) a(iv), b(i), c(ii), d(iii)

(3) a(iii), b(i), c(ii), d(iv)

(4) a(iii), b(ii), c(iv), d(i)

72. Given below are two statements one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): 0.031 has four significant figures.
Reason (R): Zeros preceding to first non-zero digit are significant.
In the light of above statements choose the correct answer.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

73. The phenomenon which proves the particle nature of light is

(1) Interference

(2) Diffraction

(3) Refraction

(4) Photoelectric effect

74. If the mass ratio of He to N_2 in a container is 2 : 1 then the ratio of their moles will be

(1) 2 : 1

(2) 4 : 1

(3) 7 : 1

(4) 14 : 1

75. If an element, X has the following isotopic composition: $^{54}X : 70\%$; $^{53}X : 20\%$; $^{55}X : 10\%$. The weighted average atomic mass of the naturally occurring element X will be

(1) 53.1 u

(2) 54.1 u

(3) 53.9 u

(4) 54.9 u

76. If 0.3 mole of $BaCl_2$ is mixed with 0.4 moles of Na_3PO_4, then maximum number of moles of $Ba_3(PO_4)_2$ and NaCl formed are respectively

(1) 0.2 and 1.6

(2) 0.3 and 1.8

(3) 0.1 and 0.6

(4) 0.4 and 2.4

77. From 220 mg of CO_2, 10^{21} molecules of CO_2 are removed. Number of remaining molecules of CO_2 will be

(1) 3.011×10^{21}

(2) 2.011×10^{21}

(3) 1.011×10^{21}

(4) 6.022×10^{21}

78. Consider the following statements
Statement I: Equal masses of S_2 and S_8 contain same number of molecules.
Statement II: Equal masses of O_2 and O_3 contain equal number of atoms.
In the light of above statements, choose the correct option.

(1) Both the statements I and II are correct

(2) Both the statements I and II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

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79. Density of NO_2 gas in g/L at STP will be

(1) 1.02

(2) 4.5

(3) 2.05

(4) 3.1

80. Suppose that A and B form the compounds B_2A_3 and B_2A. If 0.05 mole of B_2A_3 weighs 9 g and 0.1 mole of B_2A weighs 10 g, the atomic weight of A and B respectively are

(1) 30 and 40

(2) 40 and 30

(3) 20 and 5

(4) 15 and 20

81. What is the mass of glucose required to produce 44 g of CO_2, on complete combustion?

(1) 30 g

(2) 45 g

(3) 60 g

(4) 22 g

82. Volume of O_2 required at STP for complete combustion of 3 g of ethane is

(1) 6.25 L

(2) 7.84 L

(3) 8.29 L

(4) 12.32 L

83. What mass of 90% pure $CaCO_3$ will be required to neutralise 100 ml of 0.25 M HCl solution, according to the following reaction?

$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + 2H_2O(l)$

(1) 2.64 g

(2) 1.39 g

(3) 1.12 g

(4) 3.65 g

84. The mass of urea required in making 2.5 kg of 0.25 molal aqueous solution is

(1) 42 g

(2) 32 g

(3) 37 g

(4) 45 g

85. 20 mL of 1 M NaOH solution is added to 200 mL of 0.25 M NaOH solution. The final concentration of resultant solution will be

(1) 0.17 M

(2) 0.21 M

(3) 0.28 M

(4) 0.32 M

86. Consider the following statements:

Statement I: Empirical formula for both acetylene and benzene is same.

Statement II: Molecular mass of acetylene = 2.5 Molecular mass of benzene

In the light of above statements, choose the correct option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

87. The amount of zinc required to produce 1.12 ml of H_2 at STP on treatment with dilute HCl will be
(atomic mass of Zn 65.5 u)

(1) 65 g

(2) 0.065 g

(3) 32.5×10^{-4} g

(4) 6.5 g

88. An organic compound containing C, H and N gave the following analysis C = 40%; H = 13.33%; N = 46.67%. Its empirical formula would be

(1) CH_4N

(2) CH_5N

(3) $C_2H_7N_2$

(4) CH_3N

89. The total number of electrons present in orbitals having $n + l = 4$ for Cr(Z = 24) is/are

(1) 1

(2) 7

(3) 8

(4) 6

90. Match the List I with List II

	List I (Species)		List II (Number of oxygen atoms)
(a)	4.8 g of ozone gas	(i)	0.1 N_A
(b)	3.2 g of sulphur dioxide gas	(ii)	0.15 N_A
(c)	3.3 g of carbon dioxide gas	(iii)	0.2 N_A
(d)	5.6 g of carbon monoxide gas	(iv)	0.3 N_A

Choose the most appropriate answer from the following

(1) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

(2) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

(3) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(4) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

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BOTANY

91. Identify the **correct** sequence of events occurring during prophase I:

a. Chromosomes are fully condensed and the meiotic spindle is assembled.

b. Each chromosome pairs with its homologue, aligned gene by gene.

c. Dissolution of the synaptonemal complex.

d. Recombination of genetic material between the non-sister chromatids of homologous chromosomes.

Choose the **correct** option.

(1) a, b, d, c

(2) b, c, d, a

(3) b, d, c, a

(4) a, c, b, d

92. Meiosis II is called equational division because during this division,

(1) Two daughter cells are formed with equal size

(2) The number of chromosomes in the daughter cells remains the same as in the cell produced at the end of meiosis I.

(3) The number of chromosomes remains the same as that of diploid parent cell before meiosis I

(4) There is separation of homologous chromosomes.

93. Quiescent or G_0 stage of cell is

(1) Metabolically inactive stage

(2) Not seen in plant cells

(3) The stage where cells do not proliferate unless they are called to do so

(4) Part of M phase

94. A bacterium divides amitotically in 20 minutes. A suspension culture of these bacteria containing 10^3 cells/ml is grown for 160 minutes. Find out cell concentration per ml in 160 minutes.

(1) 8×10^3

(2) 10^{768}

(3) 256×10^3

(4) 160×10^3

95. The metaphase plate is

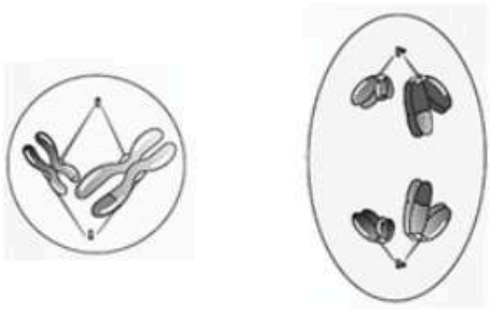
(1) Formed by the endoplasmic reticulum

(2) Expanded from one spindle pole to the other

(3) Composed of microtubules and plasma membrane

(4) An imaginary plane where the centromere of chromosomes align themselves

96. The given diagram represents two different stages A and B of a cell division.



(A) (B)

Identify the statement which can be **true** for these stages.

(1) In between stages B and A, chromosomes undergo some dispersion but do not become extremely extended form.

(2) The transition from A to B in a cell cycle is facilitated by anaphase promoting complex (APC).

(3) Just after stage A, the number of chromosomes becomes half as compared to the parent cell.

(4) Upon completion of stage A, nuclear membrane reappears.

97. Adult animal cell that does not divide is

(1) Skin cell

(2) Hair cell

(3) Stem cell

(4) Nerve cell

98. Appearance of a furrow in the plasma membrane

(1) Is observed during cytokinesis in plant cell

(2) Occurs during the formation of liquid endosperm in coconut

(3) Leads to formation of middle lamella

(4) Is observed during cytokinesis of animal cell

99. Match the following columns and choose the **correct** option.

	Column I		Column II
a.	Zygotene	(i)	Recombinase mediated activity occurs
b.	Diplotene	(ii)	Synaptonemal complex dissolves
c.	Pachytene	(iii)	Condensation of chromatin begins
d.	Leptotene	(iv)	Synapsis occurs

(1) a(ii), b(iv), c(iii), d(i)

(2) a(iv), b(i), c(ii), d(iii)

(3) a(ii), b(iii), c(iv), d(i)

(4) a(iv), b(ii), c(i), d(iii)

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100. If the male gamete of a plant species has eight chromosomes, then its parental cell will have

(1) Four bivalents during metaphase-I

(2) Eight bivalents or tetrad during pachytene

(3) Eight chromosomes during G_2 stage

(4) 32 chromosomes during G_1 stage

101. Anaphase II differs from anaphase I as the former

(1) Occurs in haploid cells

(2) Involves no division of centromere

(3) Is the sub-phase of reductional division

(4) Is not exact copy of mitosis

102. In the S phase of cell cycle

(1) Chromosome number increases

(2) Cells become haploid

(3) Amount of DNA doubles

(4) Amount of DNA becomes half

103. Continuous growth of plants throughout their life is due to

(1) Mitotic division in the apical meristem and the lateral cambium

(2) Mitotic division in primary permanent tissues

(3) Production of seeds

(4) Photosynthetically active cells only

104. Formation of interzonal fibres occur in

(1) Anaphase

(2) Metaphase

(3) Telophase

(4) Prophase

105. All of the following phenomenon may occur twice in meiosis, **except**

(1) Spindle formation

(2) Chromosome decondensation

(3) DNA duplication

(4) Centriole duplication

106. Major reorganisation of virtually all components of the cell occurs during

(1) M phase

(2) G_0 phase

(3) G_1 phase

(4) G_2 phase

107. The process that restores the diploid phase in life cycle of sexually reproducing organisms is

(1) Meiosis

(2) Fertilisation

(3) Mitosis

(4) Karyokinesis

108. Dictyotene is

(1) Movement of dictyosome within a cell

(2) Prolonged diplotene stage of oocytes of some vertebrates

(3) The final stage of prophase I

(4) A metabolic stage between telophase I and prophase II

109. Microtubules which radiate out of centrosome are called

(1) Kinetochore

(2) Asters

(3) Interzonal fibres

(4) Phragmoplast

110. G_2 phase

(1) Occurs between S and M-phase

(2) Is characterized by centriole duplication

(3) Is characterized by DNA synthesis

(4) Occurs just after G_1 phase

111. End of cell division is marked by

(1) Karyokinesis

(2) Telophase

(3) Cytokinesis

(4) Interphase

112. Yeast can progress through the cell cycle in only about

(1) 20 minutes

(2) 90 minutes

(3) 30 minutes

(4) 50 minutes

113. Cell is the fundamental, structural and functional unit of life of all living organisms as
(a) It is capable of independent existence
(b) It has various organelles to ensure independent living
(c) It can perform the essential functions of life

(1) Only (b)

(2) (b) & (c) only

(3) (c) only

(4) (a), (b) & (c)

114. Which of the following statements was given by Rudolf Virchow to modify the cell theory in its final shape?

(1) All the living cells have nucleic acid as genetic material.

(2) All cells arise from pre-existing cells.

(3) All cells are covered by an envelope

(4) All living organisms are composed of cells and products of cells.

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115. Select the **mismatched** pair w.r.t. type of cell and their shapes.

(1) Red blood cells – Round and biconcave

(2) White blood cells – Amoeboid

(3) Mesophyll cells – Spiral

(4) Nerve cells – Branched and long

116. A prokaryotic cell differs from an eukaryotic cell as the former

(1) Possesses a well-defined nucleus.

(2) Has a cell wall, as the delimiting structure.

(3) Lacks 70S type of ribosomes.

(4) Lacks membrane bound organelles.

117. Many bacteria have small circular DNA outside the genomic DNA. This extranuclear DNA

a. Confers certain unique phenotypic characters to bacteria.

b. Are used to monitor bacterial transformation with foreign DNA.

c. May attach to a single mRNA and form polysome.

d. Are also found in all eukaryotic cells.

(1) a and b

(2) b and c

(3) c and d

(4) a and d

118. Given below are two statements labelled as Assertion (A) and Reason (R).

Assertion (A): The middle layer of the cell envelope of the cells that have fimbriae, prevent these cells from bursting or collapsing.

Reason (R): The middle layer of the cell envelope in the cells that shows the presence of fimbriae, has phospholipid as major component and differs in thickness among different forms.

In the light of the above statements, select the **correct** option:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

119. Which of the following statements is **not** true for cell membrane?

(1) Present in both prokaryotic and eukaryotic cells

(2) Cell membranes possess proteins but they do not have carbohydrates

(3) Composed of lipids that are arranged in a bilayer

(4) Integral proteins are partially or totally buried in the membrane

120. Plant cells and animal cells are different as the former possess

(1) Mitochondria, cytoplasm and golgi apparatus

(2) Cell wall, plastids and a large central vacuole

(3) Centrioles, microtubules and microvilli

(4) Peroxisome, lysosome and nucleus

121. In plants, secondary cell wall is formed

(1) Inside the cell membrane

(2) Towards the outer side of the primary cell wall

(3) Between the primary cell wall and plasma membrane

(4) When the primary cell wall gets completely diminished

122. Select the **incorrect** statement regarding Golgi apparatus.

(1) This cell organelle was named by Camillo Golgi

(2) Its *cis* face is the forming face and *trans* face is the maturing face

(3) Its *cis* and *trans* faces are interconnected

(4) It consists of many cisternae stacked parallel to each other

123. Which one of the following seal the ends of chromosomes and prevent their shortening?

(1) Centromeres

(2) Kinetochores

(3) Telomeres

(4) Chromomeres

124. In animal cells, detoxification of drugs and steroid synthesis are associated to

(1) Golgi apparatus

(2) Rough endoplasmic reticulum

(3) Microbodies

(4) Smooth endoplasmic reticulum

125. Match the following columns and select the **correct** option.

	Column I		**Column II**
a.	Contractile vacuole	(i)	Stores starch
b.	RER	(ii)	Glycosylation of proteins
c.	Golgi complex	(iii)	Excretion
d.	Amyloplast	(iv)	Protein synthesis

(1) a(iii), b(i), c(ii), d(iv)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(iv), b(i), c(ii), d(iii)

(4) a(i), b(iii), c(ii), d(iv)

126. Hub in centriole is

(1) Peripheral triplet of microtubules

(2) Rod-shaped proteinaceous mass in the central part

(3) Proteinaceous amorphous pericentriolar region

(4) Radially oriented filaments

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127. Read the given statements and select the **correct** option.
Statement (A): In plants, the membrane of vacuole facilitates the transport of a numbers of ions and other materials against concentration gradients into the vacuole.
Statement (B): The concentration of ions and other materials is significantly higher in the vacuole than in the cytoplasm

(1) Only statement (A) is correct

(2) Both the statements (A) and (B) are correct

(3) Only statement (A) is incorrect

(4) Both the statements (A) and (B) are incorrect

128. Read the following statements:

A. The outer membrane and the inner membrane divide its lumen distinctly into two aqueous compartments.

B. The outer compartment is filled with a dense homogeneous substance called the matrix.

C. The matrix possesses the components required for the synthesis of proteins.

D. The matrix has single linear dsDNA and polyribosomes.

E. The infoldings of inner membrane increase the surface area for enzyme action.

Regarding mitochondria, the **false** statements are

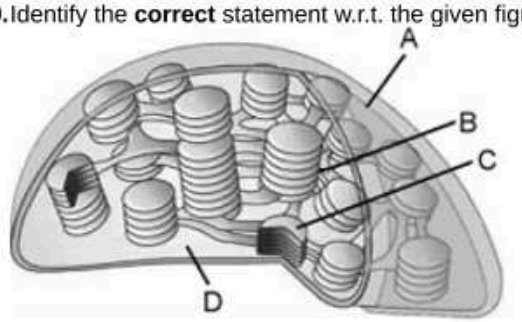
(1) A and C only

(2) A, C and D

(3) B, C and E

(4) B and D only

129. Identify the **correct** statement w.r.t. the given figure.



A. It is more permeable in comparison to the inner membrane.

B. These are stacked flat membranous tubules arranged like piles of coins which help in CO_2 fixation.

C. These are organised flattened membranous sacs present in the stroma.

D. It has enzymes for synthesis of sugar.

(1) A and B only

(2) B and C only

(3) A, C and D

(4) B, C and D

130. Satellite in chromosomes is

(1) Seen in telocentric chromosomes only

(2) A fragment beyond the secondary constriction

(3) Formed by primary constriction

(4) Devoid of DNA

131. Match Column-I with Column-II and select the **correct** option.

Column-I (Components of cell envelope)		**Column-II (Features)**
a. Slime layer	(i)	It protects the cell from loss of water and nutrients
b. Capsule	(ii)	It is similar structurally to that of eukaryotes
c. Cell wall	(iii)	It prevents the cell from bursting or collapsing
d. Cell membrane	(iv)	It allows bacterium to hide from host's immune system

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(i), b(iv), c(iii), d(ii)

(4) a(iv), b(i), c(ii), d(iii)

132. Structure found in blue-green, purple and green photosynthetic bacteria that provide buoyancy are

(1) Microtubules

(2) Centrioles

(3) Gas vacuoles

(4) Contractile vacuoles

133. Which of the following chromosomes appears L-shaped during anaphase stage of cell division?

(1) Acrocentric

(2) Telocentric

(3) Sub-metacentric

(4) Metacentric

134. Chromatin contains
(a) DNA
(b) Histone proteins
(c) RNA
(d) Non-histone proteins

(1) (a) & (b) only

(2) (a), (b) & (d) only

(3) (a), (b) & (c) only

(4) All (a), (b), (c) & (d)

135. Which of the following connects the cytoplasm of two plant cells?

(1) Polysome

(2) Plasmodesmata

(3) Middle lamella

(4) Lysosomes

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ZOOLOGY

136. Presence of both lamella and lacunae is a feature of

(1) Bone

(2) Cartilage

(3) Tendon

(4) Ligament

137. The epithelium which is made of a single thin layer of flattened cells with irregular boundaries is found in

(1) PCT of nephron

(2) Walls of blood vessels

(3) Inner surface of hollow organs like bronchioles

(4) Stomach and intestine

138. Dense irregular connective tissue having fibroblasts and mostly collagen fibres are found in

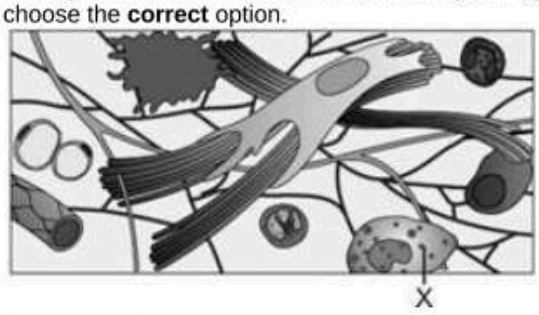
(1) Skin

(2) Ligament

(3) Tendons

(4) Blood

139. Identify the structure marked X in below given figure and choose the **correct** option.



(1) Mast cell

(2) Collagen

(3) Fibroblast

(4) Goblet cell

140. Complete the analogy.
Multicellular gland : Salivary gland : : Unicellular gland : _____
Choose the **correct** option.

(1) Goblet cell

(2) Sebaceous gland

(3) Sweat gland

(4) Gastric gland

141. All of the following structures have cartilage, **except**

(1) Tip of nose, outer ear joint

(2) Between adjacent bones of the vertebral column, limbs in adult

(3) Nasal septum, hands in adult, thoracic cage

(4) Heart, Kidney, Capillaries

142. Skeletal and cardiac muscle fibres can be differentiated on the basis of all **except**

(1) Presence of striations

(2) Branched appearance

(3) Involuntary nature

(4) Presence of communication junctions

143. Select the **correct** statements w.r.t. bones.
(a) They support and protect softer tissues and organs.
(b) Limb bones serve weight bearing functions.
(c) Their interaction with muscles having fusiform shaped fibres bring about movements.
(d) All bones are site of production of blood cells in aged individuals.
Choose the most appropriate answer from the options given below.

(1) (a) and (b)

(2) (a) and (c)

(3) (c) and (d)

(4) (b) and (d)

144. **Assertion (A):** In a cardiac muscle, when one cell receives a signal to contract, its neighbours are also stimulated to contract.
Reason (R): Communication junctions at some fusion points allow the cells to contract as a unit.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

145. All of the following statements are true for neuroglia, **except**

(1) They are the specialised cells found in brain and spinal cord

(2) More than 50% of the volume of neural tissue is composed of neuroglial cells

(3) These cells have different shapes and sizes and are thought to separate and insulate adjacent neurons.

(4) They do not support and protect the neurons

146. The type of epithelial tissue that is present in the inner lining of ducts of salivary glands is characterized by the presence of

(1) A single thin layer of flattened cells with irregular boundaries

(2) Tall and slender cells with central nuclei

(3) More than one layer of cells

(4) Cube-like cells with basal nuclei

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147. In response to stimulation, muscle fibres 'X' and therefore get 'Y'. Identify 'X' and 'Y' and select the **correct** option.

(1) **X** – Relax, **Y** – Lengthen

(2) **X** – Contract, **Y** – Shorten

(3) **X** – Contract, **Y** – Lengthen

(4) **X** – Relax, **Y** – Shorten

148. How many of the given statements is/are **correct** w.r.t. connective tissues?
a. It always has a large amount of intracellular matrix and is involved in mechanical support.
b. It is characterized by tightly packed cells that provides the transport of materials to different tissues.
c. All types of connective tissues contain fibres
d. It forms the external covering of organs and always lack fat storing cells.
Choose the **correct** option.

(1) Zero

(2) Two

(3) Three

(4) Four

149. Peripheral nucleus is characteristic feature of

(1) Cardiac muscle cells

(2) Smooth muscle cells

(3) Cuboidal epithelial cells

(4) Skeletal muscle cells

150. Consider the following carefully.
(A) Unstriated and involuntary muscle fibres present in the walls of internal organs.
(B) Contractile tissue present only in the human heart
(C) Cells in the tissue having a free surface and held together with little intercellular material.
Which of the above tissues show the presence of cell junctions?

(1) (A) and (C) only

(2) (B) and (C) only

(3) (A), (B) and (C)

(4) (A) and (B) only

151. _____facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells, for rapid transfer of ions, small molecules and sometimes big molecules.
Choose the option which fills the blank **correctly**.

(1) Tight junctions

(2) Gap junctions

(3) Adhering junctions

(4) Desmosomes

152. Fallopian tubes are lined by

(1) Ciliated epithelium

(2) Brush bordered cuboidal epithelium

(3) Squamous epithelium

(4) Pseudostratified epithelium

153. 'X' is a type of connective tissue in which orientation of fibres show a regular pattern. Select the **correct** set of examples of 'X'.

(1) Blood and lymph

(2) Bone and cartilage

(3) Tendons and ligaments

(4) Blood and bone

154. How many of the structures mentioned in the box below are included in the broad class of tissues which is most abundant and widely distributed in the body of complex animals?

Cartilage, Aerolar tissue, Blood, Adipose tissue, Bone

Choose the **correct** option.

(1) Two

(2) Three

(3) Four

(4) Five

155. Which of the following animals exhibits division of labour at the tissue level?

(1) *Hydra*

(2) *Amoeba*

(3) *Euglena*

(4) *Paramoecium*

156. An organ such as heart comprises
a. Muscular tissue b. Neural tissue
c. Connective tissue d. Epithelial tissue
Select the correct option.

(1) a & b only

(2) a only

(3) b, c and d only

(4) a, b, c & d

157. The fibres which are present in most of the connective tissue

(1) Are responsible for providing strength to the tissue

(2) Are made up of modified lipids and polysaccharides

(3) Exhibit excitability and facilitate conduction of impulses

(4) Cannot regenerate due to avascular nature of connective tissue

158. Choose the option which represents the correct decreasing order of cellular components according to their percentage of the total cellular mass.

(1) Water > Proteins > Nucleic acids > Carbohydrates > Lipids > Ions

(2) Nucleic acids > Proteins > Lipids > Carbohydrates > Ions > Water

(3) Proteins > Nucleic acids > Ions > Carbohydrates > Water > Lipids

(4) Ions > Lipids > Carbohydrates > Nucleic acids > Proteins > Water

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159. Hydroxymethyl is present as side chain in

(1) Glycine

(2) Serine

(3) Tyrosine

(4) Tryptophan

160. The _____ sugar present in RNA is ribose and the one which is present in DNA is _____.
Fill in the respective blanks with the correct option.

(1) Pentose, 3′ Deoxyribose

(2) Hexose, 3′ Deoxyribose

(3) Pentose, 2′ Deoxyribose

(4) Hexose, 2′ Deoxyribose

161. Match column I with column II and select the **correct** option.

	Column I		**Column II**
a.	Collagen	(i)	Most abundant protein in the whole of the biosphere
b.	RuBisCO	(ii)	Most abundant protein in animal world
c.	Cellulose	(iii)	Hormone that increases blood glucose levels
d.	Glucagon	(iv)	Homopolymer which is also a secondary metabolite

(1) a(ii), b(iv), c(iii), d(i)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(i), b(ii), c(iii), d(iv)

(4) a(ii), b(i), c(iv), d(iii)

162. All of the following are incorrect statements, **except**

(1) Oils have higher melting point as compared to fats.

(2) Only one heterocyclic ring is present in the structure of substituted purines.

(3) Both trypsin and collagen are heteropolymers of amino acids.

(4) Glucose is the monomeric unit of the structural polysaccharide present in the exoskeleton of arthropods.

163. Choose the **correct** statement w.r.t a competitive inhibitor.

(1) It binds irreversibly to the enzyme.

(2) Its effect can be overcome by decreasing the concentration of the enzyme.

(3) It resembles the substrate and competes with it for the active site on enzyme.

(4) It increases V_{max} of the enzyme-catalyzed reaction.

164. Adult human haemoglobin exhibits quaternary structure which contains

(1) 2α and 2δ

(2) 2α and 2β

(3) 2γ and 2δ

(4) 2α and 2γ

165. Read the following statements and choose the **correct** option.
Statement A: Lysine and valine are considered as basic amino acids as they possess an extra amino group in their structures.
Statement B: In solutions of different pH, the structure of amino acids changes due to the ionisable nature of $-NH_2$ and –COOH groups.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is incorrect

(4) Only statement B is incorrect

166. Uridine differs from uridylic acid in

(1) Absence of phosphate

(2) Presence of uracil

(3) Presence of pentose sugar

(4) Presence of phosphate

167. During the conversion of H_2O_2 into water and oxygen, haem is required, as it is

(1) A metal ion which is required for the catalytic activity of peroxidase

(2) A part of the active site of the enzyme which catalyses this conversion

(3) Transiently bound to the protein part of the enzyme involved in this reaction

(4) An inorganic compound

168. **Assertion (A):** Amino acids are substituted methanes.
Reason (R): Amino acids have four substituent groups occupying the four valency positions of the α-carbon.
In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are false

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

169. Which among the following can hold I_2 molecules in their helical portions and gives blue colour?

(1) Cellulose

(2) Starch

(3) Glycogen

(4) Glucose

170. K_m represents

(1) Maximum velocity of reaction

(2) Half the maximum velocity of reaction

(3) The temperature at which an enzyme shows optimum activity

(4) The concentration of substrate at $\frac{V_{max}}{2}$

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171. Which of the following is **incorrect** for an exothermic reaction in presence of a biological catalyst?

(1) Enzyme decreases activation energy of reaction

(2) Potential energy of product is less than that of substrate

(3) Enzyme increases velocity of reaction

(4) Enzyme substrate complex does not pass through transition state and product is formed immediately

172. If we increase the temperature by 10°C, the rate of reaction under normal conditions

(1) Will be doubled

(2) Will be tripled

(3) Remains unaffected

(4) Will be four times upto optimum temperature

173. Choose the option with **correct** sequence of events.
a. Active site of enzyme breaks the chemical bonds of the substrate.
b. The enzyme releases the product of reaction.
c. Substrate binds to active site of enzyme.
d. Binding of substrate induces enzyme to alter its shape.
e. Enzyme product complex is formed.

(1) a → b → c → d → e

(2) c → d → a → e → b

(3) c → a → d → e → b

(4) c → d → e → a → b

174. Enzymes catalysing removal of groups from substrates by mechanisms other than hydrolysis leaving double bonds are categorised under _____ of enzymes.
Choose the correct option to fill in the blank.

(1) Class II

(2) Class III

(3) Class IV

(4) Class VI

175. Cholesterol is an example of

(1) Phospholipid

(2) Simple lipid

(3) Glycolipid

(4) Derived lipid

176. If adenine makes 32% of a dsDNA molecule, then calculate the percentage of thymine, guanine and cytosine, respectively, present in this DNA molecule and select the **correct** option.

(1) 18%, 32%, 18%

(2) 18%, 18%, 32%

(3) 32%, 18%, 18%

(4) 32%, 18%, 32%

177. The **correct** representation of palmitic acid is

(1) $CH_3 - (CH_2)_{16} - COOH$

(2) $CH_3 - (CH_2)_{14} - COOH$

(3) $CH_3 - (CH_2)_7\ CH = CH\ (CH_2)_7\ COOH$

(4) $CH_3 - (CH_2)_{18} - COOH$

178. The bond present between the hydroxyl group of sugar and phosphate is

(1) Glycosidic bond

(2) Hydrogen bond

(3) Ester bond

(4) Ionic bond

179. How many of the following factors will affect the activity of a holoenzyme?
(a) Temperature
(b) pH
(c) Change in substrate concentration
(d) Absence of co-factor

(1) Four

(2) Three

(3) Two

(4) One

180. Co-enzyme nicotinamide adenine dinucleotide (NAD) contains the vitamin _____.
Choose the **correct** option for filling the blank.

(1) Thiamine

(2) Riboflavin

(3) Niacin

(4) Retinol

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